Advanced Survival Analysis of Acute Myelogenous Leukaemia (AML) Remission Data

Group 5
Faida Niyonshimiye
Oliver Nalunkuma
Sylvestre Mukiza
Patrick Makina
Kutlo Kejang

Lecturer: Prof. Innocent Ngaruye
Course: Survival Analysis

March 28, 2026

Outline

1. Introduction
2. Non-Parametric Analysis
3. Non-Parametric Hypothesis Testing
4. Cox Proportional Hazards Model
5. Model Refinement
6. Synthesis
7. Conclusions

Introduction

This project extends the parametric survival analysis of Project 1 by applying non-parametric and semi-parametric methods to the AML remission data, providing a comprehensive characterisation of survival patterns and covariate effects.

Objective: To analyse AML patient survival using non-parametric estimators, hypothesis tests and the Cox proportional hazards model, and to consolidate findings across all analytical approaches.

Research Questions:

1. What is the overall survival experience of AML patients and how does it differ across treatment, sex and FLT3 mutation level groups?
2. What does the empirical hazard trend reveal and how does it compare with the Gompertz parametric model from project 1?
3. Which covariates are significant predictors of survival after adjusting for other factors?

Dataset and Methodology

Dataset: The `myeloid` dataset from the `survival` R package – 646 AML patients in a randomised clinical trial comparing standard vs experimental treatment.

Covariates used: treatment arm (`trt`), sex and FLT3 mutation level – all baseline categorical variables. Time-dependent variables as transplant time, complete remission time and relapse time (`txtime`, `crtime`, `rltime`) were not included as covariates in the Cox model, as they represent post-baseline events rather than baseline predictors. Including them as fixed covariates would introduce immortal time bias.

Dataset and Methodology Cont.

Methods:

- **Step 1:** Kaplan-Meier estimator – overall and stratified survival curves
- **Step 2:** Nelson-Aalen estimator – cumulative hazard and kernel-smoothed hazard
- **Step 3:** Log-Rank and Wilcoxon hypothesis tests
- **Step 4:** Univariable and multivariable Cox PH models with diagnostics
- **Step 5:** Model refinement based on PH assumption assessment
- **Step 6:** Synthesis – KM vs Gompertz comparison and forest plot

Kaplan-Meier – Overall Survival Estimate

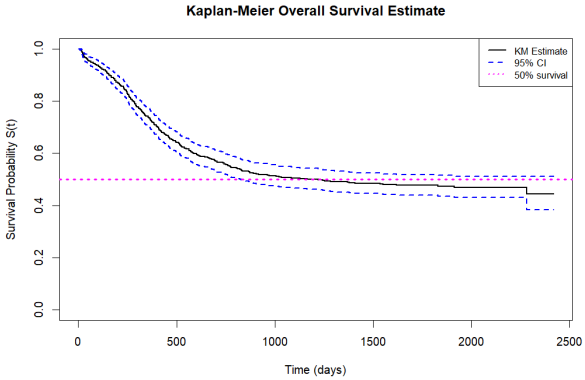

Figure 1: KM overall survival estimate with 95% CI and 50% reference line.

The KM estimate yields:

- **Median survival = 1220 days** (95% CI: 826 – not reached), meaning half the cohort survived beyond approximately 3.3 years.
- The upper confidence limit is not reached during the study period, meaning many patients survived beyond the observation time.
- The curve drops steeply in the first 500 days then gradually flattens which is consistent with the decreasing hazard identified in project 1.

Kaplan-Meier – Survival Probabilities

Table 1: KM Survival Probabilities at Key Time Percentiles.

Percentile	Time (days)	Survival Probability $S(t)$	Lower 95% CI	Upper 95% CI
25th	286.5	0.7898	0.7583	0.8226
50th	817.0	0.5407	0.5025	0.5817
75th	1733.3	0.4771	0.4386	0.5189

- At the 25th percentile of follow-up time (286.5 days), 78.98% of patients were still alive, suggesting relatively low early mortality.
- At the 50th percentile (817 days), 54.07% survived, i.e., just over half the cohort was still alive at the median follow-up.
- At the 75th percentile (1733 days), 47.71% survived. Survival stabilises after year 2, indicating that a group of patients achieves long-term remission.

Kaplan-Meier – Stratified Survival Curves

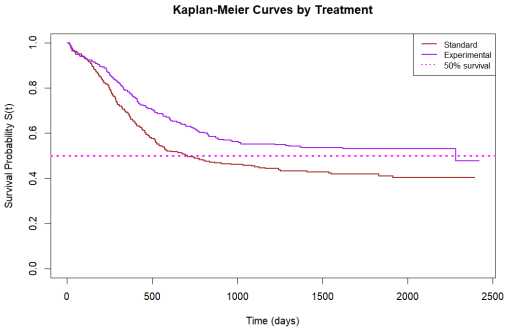

Figure 2: KM curves by treatment.

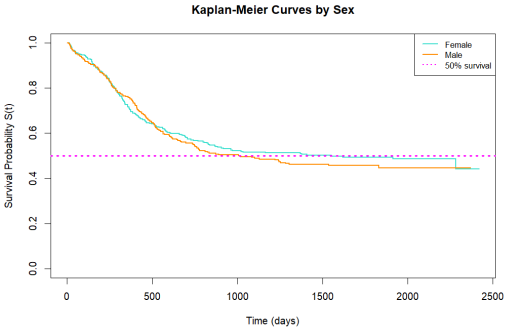

Figure 3: KM curves by sex.

Kaplan-Meier – Stratified Survival Curves Cont.

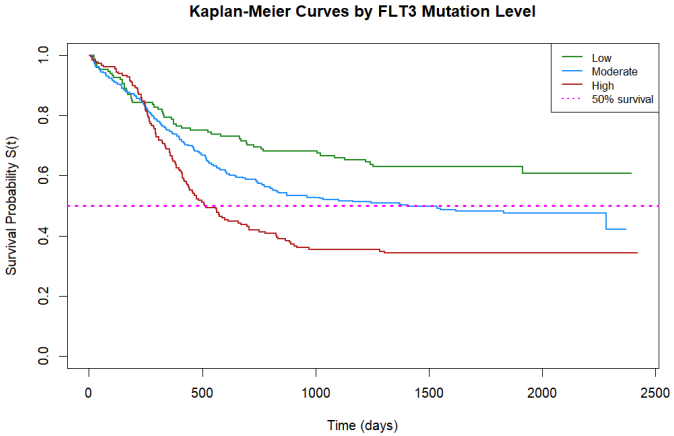

Figure 4: KM curves by FLT3 mutation level.

Nelson-Aalen – Cumulative Hazard

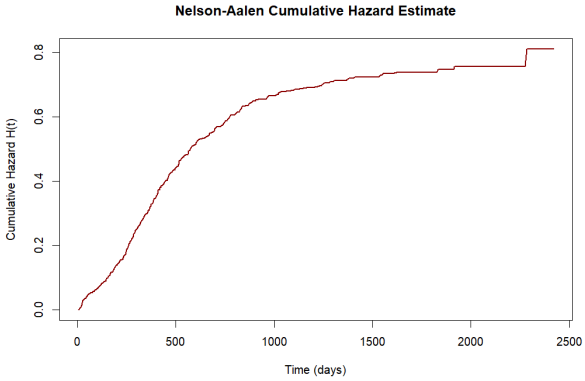

Figure 5: Nelson-Aalen cumulative hazard estimate $\hat{H}(t)$.

- $\hat{H}(t)$ increases steeply and concavely in the first 500 days, then flattens, i.e., the rate of accumulation of hazard slows down over time.
- A concave cumulative hazard shows a decreasing hazard function over time, which is consistent with the Gompertz model from project 1 ($\hat{\gamma} < 0$).
- By day 1095, $\hat{H}(1095) = 0.6813$, meaning the cumulative risk accumulated over 3 years corresponds to roughly 68 expected deaths per 100 patients.

Nelson-Aalen – Estimates Hazard Comparison

Table 2: Nelson-Aalen Cumulative Hazard and Hazard vs Gompertz.

Time (days)	**Cumulative Hazard $\hat{H}(t)$**	**Non-Parametric Hazard $h(t)$**	**Gompertz Hazard $h(t)$**
30	0.0305	0.001017	0.001124
90	0.0625	0.000532	0.001029
180	0.1210	0.000650	0.000902
365	0.3130	0.001038	0.000687
730	0.5716	0.000709	0.000402
1095	0.6813	0.000300	0.000235

Nelson-Aalen – Hazard Comparison

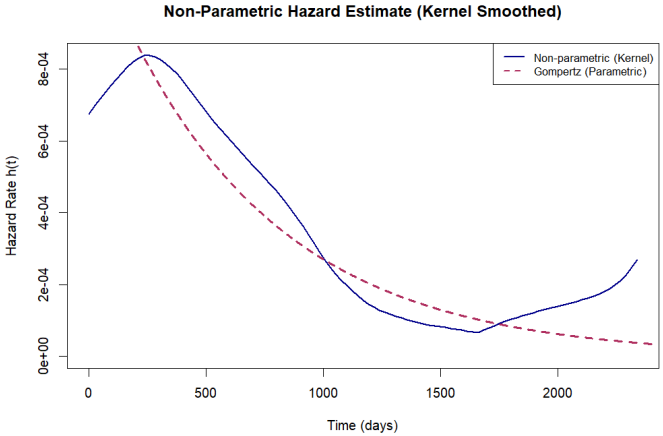

Figure 6: Kernel-smoothed non-parametric hazard vs Gompertz parametric hazard.

Nelson-Aalen – Hazard Trend Interpretation

Looking at the kernel-smoothed hazard plot and Table 2:

- The **non-parametric hazard** shows an initial rise to a peak around day 200-300, then a broad decline, which indicates a non-monotone pattern, unlike the purely decreasing Gompertz hazard.
- The **Gompertz hazard** assumes a smooth, monotonically decreasing hazard over time. It captures the overall declining trend well but misses the early peak observed empirically.
- This discrepancy is clinically meaningful as the early peak in the non-parametric hazard reflects the acute burden of treatment toxicity, infection and disease relapse in the first months after treatment.
- Despite this difference, both curves converge and track closely beyond day 500, suggesting the Gompertz model is a reasonable approximation over the longer term even if it smooths over the early dynamics.

Log-Rank and Wilcoxon Tests

Table 3: Log-Rank and Wilcoxon Test Results.

	Log-Rank		Wilcoxon	
Covariate	**Chi-Square**	**p-value**	**Chi-Square**	**p-value**
Treatment	9.590	0.0020	10.095	0.0015
Sex	0.462	0.4969	0.197	0.6568
FLT3 Mutation Level	24.134	<0.001	19.202	<0.001

- **Treatment and FLT3** show statistically significant differences in survival under both tests (p-value < 0.05), meaning that survival distributions differ meaningfully across groups.
- **Sex** shows no significant difference under either test (p-value > 0.05).
- Both tests yield consistent conclusions for all three covariates. Since the Wilcoxon test weights early events more heavily, consistency confirms that the survival differences are not confined to the early period alone.

Cox PH Model – Univariable Results

Table 4: Univariable Cox PH Model Results.

Covariate	Hazard Ratio	Lower 95%	Upper 95%	p-value
Treatment: Experimental vs Standard	0.708	0.568	0.882	0.0021
Sex: Male vs Female	1.079	0.866	1.344	0.4978
FLT3 Mutation: Moderate vs Low	1.506	1.104	2.056	0.0098
FLT3 Mutation: High vs Low	2.184	1.579	3.020	<0.001

Experimental treatment reduces the hazard of death by approximately 29% compared to standard (HR $= 0.708$). High FLT3 mutation more than doubles the hazard of death compared to low FLT3 mutation (HR $= 2.184$). Sex was not a statistically significant predictor of the hazard of death (HR $= 1.079$, p-value $= 0.498$).

Cox PH Model – Multivariable Results

Table 5: Multivariable Cox PH Model Results.

Covariate	**HR**	**Lower 95%**	**Upper 95%**	**p-value**
Treatment: Experimental vs Standard	0.688	0.551	0.859	<0.001
Sex: Male vs Female	1.177	0.942	1.469	0.1516
FLT3 Mutation: Moderate vs Low	1.524	1.116	2.080	0.0080
FLT3 Mutation: High vs Low	2.240	1.618	3.103	<0.001

After adjusting for all other covariates, experimental treatment reduces the hazard of death by approximately 31% compared to standard treatment (HR = 0.688, 95% CI: 0.551–0.859, p-value $<$ 0.001). High and moderate FLT3 mutation levels increase the hazard of death by 124% and 52% respectively compared to low FLT3 (HR = 2.240, p-value $<$ 0.001; HR = 1.524, p-value = 0.008). Sex remains a non-significant predictor after adjustment (HR = 1.177, p-value = 0.152). Hazard ratios change only marginally from univariable estimates, suggesting minimal confounding between covariates.

Cox PH Model – Proportional Hazards Assumption

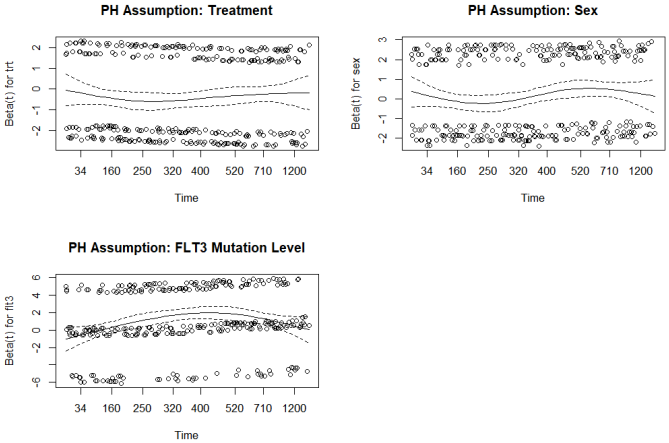

Figure 7: Schoenfeld residuals plots for each covariate.

Cox PH Model – Proportional Hazards Assumption Cont.

Schoenfeld residuals test results:

- Treatment: p-value $= 0.548$
- Sex: p-value $= 0.178$
- FLT3: p-value $= 0.089$
- Global: p-value $= 0.118$

All p-value > 0.05 meaning that the PH assumption **holds for all covariates** and globally. The Schoenfeld residual plots show no systematic trend over time, confirming proportionality. **No model refinement is required.**

Cox PH Model – Influential Observations

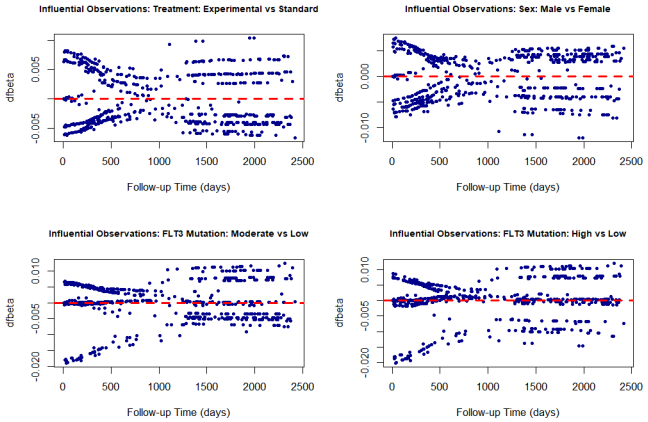

Figure 8: dfbeta residual plots showing the influence of individual observations on the Cox model coefficient estimates.

Cox PH Model – Martingale Residuals

Martingale residuals are used to assess whether continuous covariates have a linear relationship with the log-hazard.

Why it was not applied here:

- All covariates in the fitted model – treatment, sex and FLT3 mutation level – are categorical. The linearity assumption does not apply to categorical variables.
- The time variables (`txtime`, `crtime`, `rltime`) were excluded from the model, as they are post-baseline event times and not appropriate as fixed covariates.

Conclusion: The Martingale residual check is **not applicable** in this analysis, as there are no continuous baseline covariates requiring assessment.

Model Refinement

The Schoenfeld residuals test confirmed that the proportional hazards assumption holds for all covariates (all p-value > 0.05) and globally (p-value $= 0.118$).

Since no violations were detected:

- **Stratification** is not required – no covariate needs to be removed from the linear predictor.
- **Time-dependent coefficients** are not required – no covariate exhibits a time-varying effect.
- **Spline terms** are not required – all covariates are categorical, so the linearity assumption in the log-hazard does not apply.

The **multivariable Cox model** (`SurvObj ~ trt + sex + flt3`) is therefore confirmed as the **final model** with no further refinement needed.

KM vs Gompertz Survival Curve

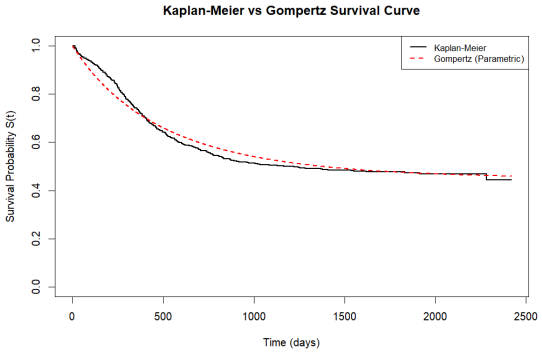

Figure 9: Kaplan–Meier vs Gompertz survival curves.

- The Gompertz parametric curve **closely tracks** the KM empirical curve throughout the observation window, confirming the adequacy of the fitted Gompertz model.
- The main divergence occurs in the **first 300 days** where the KM curve drops more steeply early on, reflecting the early mortality peak seen in the non-parametric hazard that the Gompertz model smooths over (Figure 6).
- Beyond day 500 the two curves are virtually indistinguishable, supporting the use of the Gompertz model for long-term survival estimation.

Forest Plot: Final Cox Model

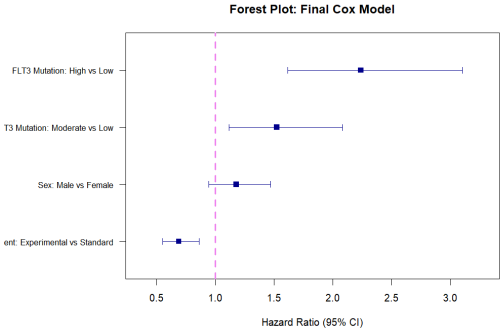

Figure 10: Forest plot of hazard ratios from the final multivariable Cox model.

Experimental treatment (HR = 0.688) and **high FLT3 mutation** (HR = 2.240) are the strongest predictors of survival. Moderate FLT3 (HR = 1.524) is also significant. Sex is not a significant predictor, with its confidence interval crossing the null value of HR = 1.

Conclusions

- The KM estimate shows a median survival of 1220 days. Survival differs significantly by treatment and FLT3 mutation level but not by sex. The experimental treatment and low FLT3 mutation are associated with better outcomes.
- The empirical hazard shows a non-monotone pattern with an early peak around day 200–300 followed by a broad decline. The Gompertz model captures the overall declining trend but smooths over this early peak. Both approaches agree on the long-term pattern.
- Experimental treatment (HR $= 0.688$, p-value < 0.001) and high FLT3 mutation level (HR $= 2.240$, p-value < 0.001) are the key independent predictors of survival after adjusting for other covariates.

Recommendations

- **Prioritise early clinical monitoring** – the hazard peak in the first year confirms that the critical window for intervention is the early post-treatment period.
- **Use FLT3 mutation level** as a stratification factor in future trials – high FLT3 patients have more than double the hazard of death and may require more aggressive treatment protocols.
- **Consider time-dependent covariate modelling** using `tmerge()` to properly incorporate transplant, remission and relapse timing in future analyses as these are post-baseline events.

References

Le-Rademacher, J. G., Peterson, R. A., Therneau, T. M., Sanford, B. L., Stone, R. M., & Mandrekar, S. J. (2018). Application of multi-state models in cancer clinical trials. *Clinical Trials*, 15(5), 489–498.

QnA

Thank You!

Questions and comments are welcome.

https://github.com/patrickmakina/Group-5-Survival-Analysis.git